Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/09/2019
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
PI Organization Type:	UNIVERSITY	**Phone:**	4408295283
Organization Name:	Case Western Reserve University		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	620 Glennan Hall, 10900 Euclid Avenue		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	03/01/2016	**End Date:**	02/28/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX16AE84A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A. Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a		

Task Description:	representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder and wake regions result in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with several embedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	One-g to micro-g drop tests of PMMA spheres under different degrees of preheating and forced air flow has been performed. Embedded thermocouples enable the quantification of the degree of subsurface preheating. A quenched (within the 5-s drop duration) vs non-quenched corridor has been identified in a map with forced velocity and solid sub-surface temperature gradient as coordinates. The transient flame response in this step g-decrease has been found to be highly dynamic with multi-dimensional features. A corresponding numerical model has been formulated and solved by our colleagues, Professor Sheng-Yen Hsu and graduate student, Ching-Wei Wu, which provided additional details of the transient flame and solid responses.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Urban DL, Ferkul PV, Olson SL, Ruff GA, Easton J, T'ien JS, Liao Y-TT, Li C, Fernandez-Pello AC, Torero JL, Legros G, Eigenbrod C, Smirnow N, Fujita O, Rouvreau S, Toth B, Jomaas G. "Flame spread: Effect of microgravity and scale." Combustion and Flame. 2019 Jan;199:168-82. https://doi.org/10.1016/j.combustflame.2018.10.012 , Jan-2019
Articles in Peer-reviewed Journals	Endo M, T'ien JS, Ferkul PV, Olson SL, Johnston MC. "Flame growth around a spherical solid fuel in low speed forced flow in microgravity." Fire Technology. First Online: 23 April 2019. https://doi.org/10.1007/s10694-019-00848-2 , Apr-2019
Papers from Meeting Proceedings	Shang W, Johnston M, Capetillo J, T'ien J. "Laboratory scale testing of Nomex fabric as a protective flame barrier in enriched oxygen atmosphere." 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. Paper 2 D05. , Mar-2019